BATCH NO:MP-AIDS-05

CLIMATE ANALYZER: DEEP LEARNING FOR VEGETATION STRESS DETECTION AND SHORT-TERM RISK FORECASTING

Major project report submitted
in partial fulfillment of the requirement for award of the degree of

Bachelor of Technology
in
Artificial Intelligence and Data Science

By

M. SABREEN ASHIKA (22UEAD0057) **(VTU 21346)**
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Under the guidance of
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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
SCHOOL OF COMPUTING

VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE AND TECHNOLOGY

(Deemed to be University Estd u/s 3 of UGC Act, 1956)
Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA

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CERTIFICATE

It is certified that the work contained in the project report titled "CLIMATE ANALYZER: DEEP LEARNING FOR VEGETATION STRESS DETECTION AND SHORT-TERM RISK FORECASTING" by "M. SABREEN ASHIKA & (22UEAD0057), B. MONIKKA & (22UEAD0036)" has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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DECLARATION

We declare that this written submission represents my ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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APPROVAL SHEET

This project report entitled CLIMATE ANALYZER: DEEP LEARNING FOR VEGETATION STRESS DETECTION AND SHORT-TERM RISK FORECASTING by M. SABREEN ASHIKA (22UEAD0057), B. MONIKKA (22UEAD0036) is approved for the degree of B.Tech in Artificial Intelligence and Data Science.

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ABSTRACT

Climate change is altering rainfall patterns, temperature regimes, and land cover, posing growing risks to agriculture, disaster management, and resource planning. Timely detection of vegetation stress and short-term climate anomalies is crucial for informed decision-making. This paper presents a software-only framework that fuses satellite imagery and climate data using lightweight deep learning models. Sentinel-2 Level-2A imagery is processed by a Convolutional Neural Network (CNN) to generate vegetation stress maps, with labels automatically derived from Normalized Difference Vegetation Index (NDVI) deviations relative to a three-year seasonal baseline. Simultaneously, an LSTM (Long Short-Term Memory) network forecasts ERA5-Land rainfall and temperature one to three months ahead. The outputs are fused into low/medium/high risk maps, trend charts, and alert messages. Experiments demonstrate that the system reliably identifies stressed regions and produces actionable forecasts while using only publicly available data and efficient models. The framework is adaptable across regions and requires no manual annotation, enabling rapid deployment. The main contributions include: (i) a CNN-LSTM fusion approach for integrating NDVI and climate data, (ii) automatic global-scale vegetation stress detection, and (iii) multi-level risk classification validated with Sentinel-2 and ERA5 data. This system supports farmers, planners, and local authorities in making faster, better-informed decisions under climate-related threats.

Keywords: Climate Change, Sentinel-2, Convolutional Neural Networks (CNN), Normalized Difference Vegetation Index (NDVI), ERA5-Land, Long Short-Term Memory (LSTM), Risk Mapping

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LIST OF ACRONYMS AND ABBREVIATIONS

AOI	Area of Interest
CNN	Convolutional Neural Network
ERA5	ECMWF Reanalysis v5
FCN	Fully Convolutional Network
IoU	Intersection over Union
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
NDVI	Normalized Difference Vegetation Index
NIR	Near-Infrared
RMSE	Root Mean Square Error
SAR	Synthetic Aperture Radar
SDG	Sustainable Development Goal

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Chapter 1

INTRODUCTION

1.1 Introduction

Climate change has become one of the most critical challenges of our time, affecting rainfall patterns, temperature regimes, and land cover across the globe. Agriculture, disaster management, and urban planning are all deeply impacted by these rapid environmental shifts. One of the most pressing needs in this context is the ability to detect vegetation stress early and forecast near-term climatic risks, enabling informed decisions by farmers, planners, and local authorities.

Satellite remote sensing has made it possible to monitor vegetation health over large areas at regular intervals. Sentinel-2 Level-2A imagery, provided freely by the European Space Agency, offers high-resolution multispectral data at a 10 m spatial resolution and a revisit time of five days. The Normalized Difference Vegetation Index (NDVI), computed from the Red (B4) and Near-Infrared (B8) bands, is a widely used and robust indicator of vegetation health. When NDVI drops significantly below its seasonal baseline, it signals stress, often attributable to drought, heat, or pest activity.

Simultaneously, climate reanalysis datasets such as ERA5-Land, maintained by the European Centre for Medium-Range Weather Forecasts (ECMWF), provide consistent long-term monthly records of rainfall and temperature. These datasets are used to understand anomalies and to train predictive models for short-term climate forecasting.

Deep learning has emerged as a powerful tool for extracting patterns from complex, high-dimensional data. Convolutional Neural Networks (CNNs) are particularly effective for spatial image analysis, while Long Short-Term Memory (LSTM) networks excel at learning from temporal sequences. This project, Cli-

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mate Analyzer, integrates both approaches into a single end-to-end framework. A lightweight CNN processes Sentinel-2 image patches to produce pixel-level vegetation stress probability maps. An LSTM model takes the most recent 36 months of ERA5-Land climate data to forecast rainfall and temperature one to three months into the future.

The two model outputs are fused using a simple, interpretable rule to generate risk maps classified as low, medium, or high. These maps are accompanied by trend charts and alert messages that communicate risk clearly to end users. The entire framework relies only on publicly available data, uses efficient lightweight models, and requires no manual labelling — making it suitable for rapid deployment across different geographic regions.

1.2 Background

Vegetation stress monitoring using satellite data has a long history. The NDVI, introduced by Tucker (1979), remains one of the simplest and most effective measures of vegetation health. Its deviation from a seasonal baseline provides a reliable signal for detecting drought stress, crop loss, and ecosystem degradation. With the launch of the Sentinel-2 constellation, it became possible to compute NDVI at field-scale resolution with high frequency across the globe.

Climate reanalysis products have made it practical to study historical climate patterns and to build data-driven forecast models. ERA5 and ERA5-Land, produced by ECMWF, provide globally consistent records of atmospheric variables from 1940 to the present. These datasets are validated against observations and are widely used in both academic research and operational climate services.

Deep learning for remote sensing has advanced rapidly in recent years. U-Net style architectures, originally developed for biomedical image segmentation, have been successfully applied to vegetation mapping, land cover classification, and crop monitoring. LSTM networks, introduced by Hochreiter and Schmidhuber (1997), have proven effective for time series forecasting in hydrology, climate science, and agricultural applications. Hybrid CNN-LSTM models that combine spatial and tem-

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poral reasoning have shown further improvements over single-modality baselines, motivating the design of this project.

1.3 Objective

The primary objective of this project is to develop a software-only, end-to-end deep learning framework that integrates satellite imagery and climate time series for vegetation stress detection and short-term climate risk forecasting. The system is designed to be lightweight, reproducible, and deployable across diverse geographic regions without requiring manual annotation.

Specific objectives include: (i) processing Sentinel-2 Level-2A multispectral imagery to compute NDVI and automatically generate pixel-level stress labels using a three-year seasonal baseline; (ii) training a lightweight CNN (U-Net/FCN style) to predict per-pixel vegetation stress probability from image patches; (iii) training an LSTM network on ERA5-Land monthly climate series to forecast rainfall and temperature one to three months ahead; (iv) fusing the CNN stress maps and LSTM climate forecasts through a transparent rule-based approach to produce low, medium, and high risk maps; and (v) generating actionable outputs including risk maps, NDVI change visualizations, trend charts, and alert messages that can support agricultural and planning decisions.

1.4 Problem Statement

Despite the abundance of satellite imagery and climate reanalysis data, extracting timely, actionable risk information from these sources remains challenging. Most existing tools focus either on vegetation indices alone or on climate time series in isolation, limiting their ability to produce comprehensive and reliable early warnings. Manual annotation for training supervised models is costly and impractical at a global scale. Furthermore, many deep learning approaches for this domain rely on heavy architectures, large proprietary datasets, and complex preprocessing pipelines that limit reproducibility and adaptability.

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This project addresses these gaps by proposing a fully automated, self-supervised framework that uses NDVI anomalies to generate training labels without human annotation. By combining a lightweight CNN for spatial stress mapping with an LSTM for temporal climate forecasting, and fusing both outputs into a clear risk product, the system delivers actionable information that neither modality alone can provide. The framework is designed to run monthly on modest hardware, using only freely available data, and produces outputs that are easy to interpret for non-specialist users such as farmers, extension workers, and local planners.

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Chapter 2

LITERATURE REVIEW

[1] M. Drusch et al., presented the Sentinel-2 mission, which provides high-resolution optical satellite imagery for land monitoring and environmental analysis. The study highlights the importance of multispectral data in observing vegetation conditions and land cover changes. Sentinel-2 data has become widely used for agricultural monitoring due to its high spatial resolution and frequent revisit time, enabling effective large-scale vegetation analysis.

[2] J. Muñoz-Sabater et al., introduced the ERA5-Land dataset, which provides long-term and consistent climate reanalysis data, including rainfall and temperature. This dataset is widely used for climate studies and environmental monitoring. The study emphasizes the reliability of ERA5-Land data in analyzing climate variability and supporting forecasting applications.

[3] C. J. Tucker proposed the Normalized Difference Vegetation Index (NDVI), which is a widely used indicator for measuring vegetation health. NDVI is calculated using near-infrared and red spectral bands and helps in identifying stressed vegetation, drought conditions, and seasonal changes. This method has been extensively applied in remote sensing for agricultural and environmental monitoring.

[4] Recent studies have utilized deep learning techniques such as Convolutional Neural Networks (CNNs) for analyzing satellite imagery. CNN-based models have shown strong performance in detecting vegetation patterns, land cover classification, and stress identification due to their ability to extract spatial features from multispectral images.

[5] Several researchers have applied Long Short-Term Memory (LSTM) networks for time-series forecasting of environmental variables such as rainfall, temperature, and vegetation indices. LSTM models are capable of capturing temporal dependencies and have been successfully used in climate prediction and drought forecasting.

[6] Hybrid models combining CNN and LSTM have been proposed in recent

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works to improve prediction accuracy. These models integrate spatial and temporal information, providing better results compared to individual models. Studies have shown that combining satellite imagery with climate data enhances the ability to detect vegetation stress and predict future risks.

[7] Although existing approaches provide valuable insights, many systems focus either on vegetation monitoring or climate forecasting independently. There is limited research on integrating both aspects into a unified framework that provides actionable risk predictions.

From the above literature, it is evident that combining satellite-based vegetation analysis with climate forecasting using deep learning techniques can significantly improve the accuracy and effectiveness of environmental monitoring systems. This motivates the development of the proposed system, which integrates CNN and LSTM models for vegetation stress detection and short-term risk forecasting.

2.1 Existing System

Existing vegetation stress monitoring systems typically rely on threshold-based NDVI analysis applied independently to individual images. These systems compute NDVI from satellite data and flag pixels that fall below a fixed value as stressed, without accounting for seasonal variation or local land cover characteristics. Such static approaches often produce high false-alarm rates during natural seasonal transitions and miss subtle multi-month trends that precede full-scale crop loss.

On the climate forecasting side, existing operational systems use statistical models or general circulation models (GCMs) that are computationally expensive, require specialized inputs such as soil moisture and land-use layers, and are difficult to deploy rapidly for a specific area of interest. The combination of vegetation stress information and climate forecasts into a unified, interpretable risk product is not readily available in lightweight, open-access tools.

2.2 Related Work

Bartold et al. (2024) examined the sensitivity of satellite-derived vegetation indices to plant drought stress in grasslands, finding that NDVI deviations from

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seasonal baselines are strong indicators of early stress onset. Gupta and Gaurav (2024) proposed a hybrid stacked model for drought prediction through rainfall forecasting, demonstrating improved accuracy over single-model approaches when combining multiple data sources.

Gao et al. (2023) introduced an NDVI forecasting model combining time series decomposition with a CNN-LSTM architecture, achieving improved accuracy over individual models. Fathollahi et al. (2024) developed a global NDVI forecasting model from air temperature, soil moisture, and precipitation using a deep neural network. Shams Eddin and Gall (2024) demonstrated that the Focal-TSMP regional climate simulation framework can predict vegetation health and assess agricultural drought. These works collectively motivate a CNN-LSTM fusion approach that operates directly on public satellite and reanalysis data.

2.3 Research Gap

Despite significant progress in both vegetation monitoring and climate forecasting, several gaps remain. Most existing CNN-based vegetation stress models require manual annotation or additional auxiliary inputs such as soil moisture data, limiting their scalability. Many LSTM-based climate forecasting systems are evaluated independently, without integration into spatial risk products. The combination of self-supervised NDVI labelling, pixel-level CNN stress mapping, and short-term LSTM climate forecasting into a single unified and reproducible pipeline has not been widely demonstrated on publicly available data alone.

This project addresses these gaps by providing a fully automatic framework that: uses NDVI anomalies relative to a three-year baseline as self-generated labels; trains a lightweight CNN for pixel-level stress classification; combines the spatial stress fraction with LSTM-forecasted climate anomalies through a transparent fusion rule; and produces actionable low/medium/high risk maps without requiring manual annotation, proprietary data, or heavy computational resources.

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Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

Current vegetation monitoring systems largely depend on static NDVI thresholds applied to individual satellite images. These systems identify stressed vegetation by comparing current NDVI values against fixed global cutoffs, ignoring the seasonal and regional variability of vegetation cycles. As a result, regions undergoing normal dry-season senescence are often misclassified as stressed, while early-stage stress events that do not yet meet the threshold are missed entirely.

Existing climate risk assessment tools are typically designed for expert use and require proprietary climate model outputs, high-performance computing infrastructure, and domain-specific calibration. They do not integrate satellite vegetation data with climate forecasts in an automated, transparent pipeline. The lack of integrated, lightweight tools accessible to non-specialists — including farmers, extension officers, and local planners — represents a major gap in early warning capabilities for climate-related agricultural risk.

Disadvantages of Existing System:

- Static NDVI thresholds do not account for seasonal or regional variation, leading to high false-alarm and miss rates.
- No integration between spatial vegetation stress maps and short-term climate forecasts.
- Dependence on proprietary data, expensive software, and high-performance computing.
- Require manual annotation for training supervised models, which is not scalable.
- Outputs are typically not interpretable for non-specialist end users.

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3.2 Proposed System

The proposed Climate Analyzer system is a software-only, end-to-end deep learning framework that integrates Sentinel-2 satellite imagery with ERA5-Land climate reanalysis data. Vegetation stress labels are automatically derived by computing the NDVI anomaly relative to a three-year same-month baseline, eliminating the need for manual annotation. A lightweight CNN (U-Net/FCN style) is trained to produce per-pixel stress probability maps from 64x64 multispectral patches. An LSTM network is trained on 36 months of ERA5-Land rainfall and temperature data to forecast the next one to three months. The CNN stress fraction and LSTM climate anomalies are fused through a transparent rule to assign low, medium, or high risk to the area of interest.

The system produces color-coded risk maps, before-and-after NDVI change views, trend charts for rainfall and temperature forecasts, and short alert messages. All components use publicly available data and modest computational resources, making the system reproducible and adaptable to new regions without retraining from scratch.

Advantages of Proposed System:

- Self-supervised labeling from NDVI baselines eliminates manual annotation.
- CNN and LSTM fusion provides both spatial and temporal risk information.
- Lightweight architecture runs on CPU or standard GPU hardware.
- Uses only freely available Sentinel-2 and ERA5-Land data.
- Produces interpretable, actionable outputs suitable for farmers and planners.
- Fully reproducible with time-based data splits and logged hyperparameters.

3.3 Feasibility Study

A feasibility study was conducted to evaluate the viability of the Climate Analyzer system from economic, technical, and social perspectives. The system was assessed against available data sources, computational requirements, open-source tooling, and

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the practical needs of target end users. The results confirm that the proposed system is feasible for development, deployment, and ongoing use.

3.3.1 Economic Feasibility

The Climate Analyzer relies entirely on publicly available, freely downloadable datasets. Sentinel-2 Level-2A imagery is distributed freely by the European Space Agency through the Copernicus Open Access Hub, and ERA5-Land monthly data is available without charge from the ECMWF Climate Data Store. All software components — including Python, PyTorch, NumPy, Pandas, Matplotlib, xarray, and netCDF4 — are open-source with no licensing cost. Model training and inference can be performed on standard hardware without the need for dedicated high-performance computing infrastructure. The absence of proprietary data, licensed software, or cloud compute costs makes this system highly cost-effective for both development and long-term operation.

3.3.2 Technical Feasibility

The technical stack is well-established and widely supported. PyTorch provides a flexible and efficient platform for implementing both the CNN and LSTM models. The CNN adopts a lightweight U-Net-inspired architecture with two encoder levels and one transposed convolution for upsampling, keeping parameter count low while retaining sufficient representational capacity for binary stress classification. The LSTM model uses 64 hidden units and 2 layers, processing 36-month input sequences to generate 3-month forecasts, which is computationally efficient. The full monthly pipeline — including data loading, preprocessing, model inference, and output generation — runs in approximately 25 minutes on a modest workstation. All modules are implemented in Python and can be run in Google Colab or on any standard machine with at least 8 GB RAM.

3.3.3 Social Feasibility

The Climate Analyzer addresses a real and growing societal need. Climate change is increasing the frequency and intensity of drought events, threatening food security, water availability, and livelihoods in agricultural communities worldwide. By pro-

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viding automated, region-agnostic vegetation stress maps and short-term climate risk forecasts, the system can support farmers in making timely irrigation and crop management decisions, enable local planners to allocate water resources proactively, and help disaster management agencies identify at-risk areas before conditions worsen. The system's reliance on open data and its output in the form of simple maps, charts, and alert messages ensures that it can be adopted by users with limited technical expertise.

3.4 System Specification

3.4.1 Tools and Technologies Used

- **Programming Language:** Python 3.10+
- **Deep Learning Framework:** PyTorch 2.0+
- **Data Processing:** NumPy 2.0, Pandas 2.2, xarray 2025.11, netCDF4 1.7.3
- **Visualization:** Matplotlib 3.10
- **Satellite Data:** Sentinel-2 Level-2A (Copernicus Open Access Hub)
- **Climate Data:** ERA5-Land Monthly (ECMWF Climate Data Store)
- **Development Environment:** Google Colab , Jupyter Notebook
- **Version Control:** GitHub

3.4.2 Standards and Policies

Python (Google Colab / Jupyter Notebook)
Python is a high-level, general-purpose programming language used throughout this project for data processing, model training, and result generation. Google Colab provides a cloud-based Jupyter notebook environment with free access to GPU resources, enabling efficient model training without local hardware constraints. Jupyter Notebook facilitates interactive development and inline visualization of results.
Standard Used: ISO/IEC 14882 (Programming Language Standards)

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PyTorch

PyTorch is an open-source deep learning framework developed by Meta AI Research. It provides dynamic computational graphs, GPU acceleration, and a rich ecosystem of tools for building neural network architectures. In this project, PyTorch is used to implement both the CNN stress mapper (TinyUNet) and the LSTM forecaster (ClimateLSTM).

Standard Used: ISO/IEC 25010 (Software Quality)

Sentinel-2 / ERA5-Land Data

Data used in this project are distributed under open-access licenses by the European Space Agency and ECMWF respectively. Sentinel-2 products are subject to the Copernicus Data Licence, and ERA5-Land data is subject to the Copernicus Climate Change Service (C3S) Licence.

Standard Used: ISO/IEC 27001 (Information Security Management)

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Chapter 4

SYSTEM DESIGN AND METHODOLOGY

4.1 System Architecture

The Climate Analyzer system is organized as a sequential processing pipeline. Raw satellite imagery (Sentinel-2 L2A) and climate time series (ERA5-Land) are ingested and preprocessed to a common spatial and temporal grid. NDVI is computed monthly, and a three-year same-month median baseline is used to generate automatic stress labels. These labels train a lightweight CNN (TinyUNet) to produce per-pixel stress probability maps. Simultaneously, an LSTM (ClimateLSTM) is trained on 36-month ERA5-Land sequences to forecast rainfall and temperature up to three months ahead. The outputs of both models are fused through a risk classification rule to produce the final risk maps, trend charts, and alert messages.

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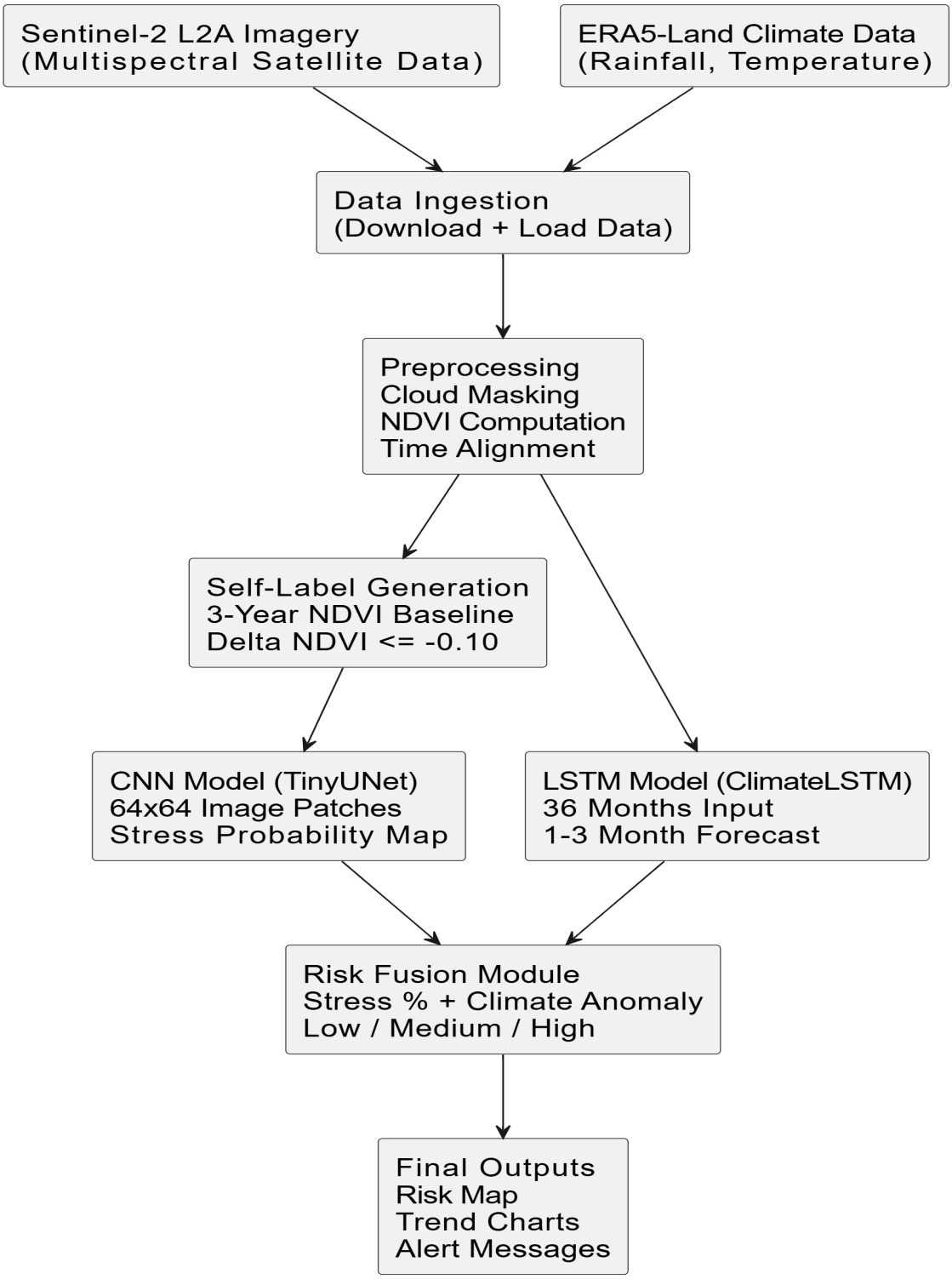

Figure 4.1: **System Architecture of Climate Analyzer**

4.2 Design Phase

4.2.1 Data Flow Diagram

The data flow begins with the ingestion of Sentinel-2 L2A tiles and ERA5-Land monthly series for the area of interest. Images are cloud-masked, reprojected, and tiled into 64x64 patches. NDVI is computed per patch per month. The three-year NDVI baseline is subtracted to yield ΔNDVI, which drives the self-label generator. The CNN receives multi-band patches and outputs stress probability maps. ERA5-Land time series are windowed into 36-month sequences that feed the LSTM. Both outputs converge in the fusion module, which assigns risk levels and generates the final products.

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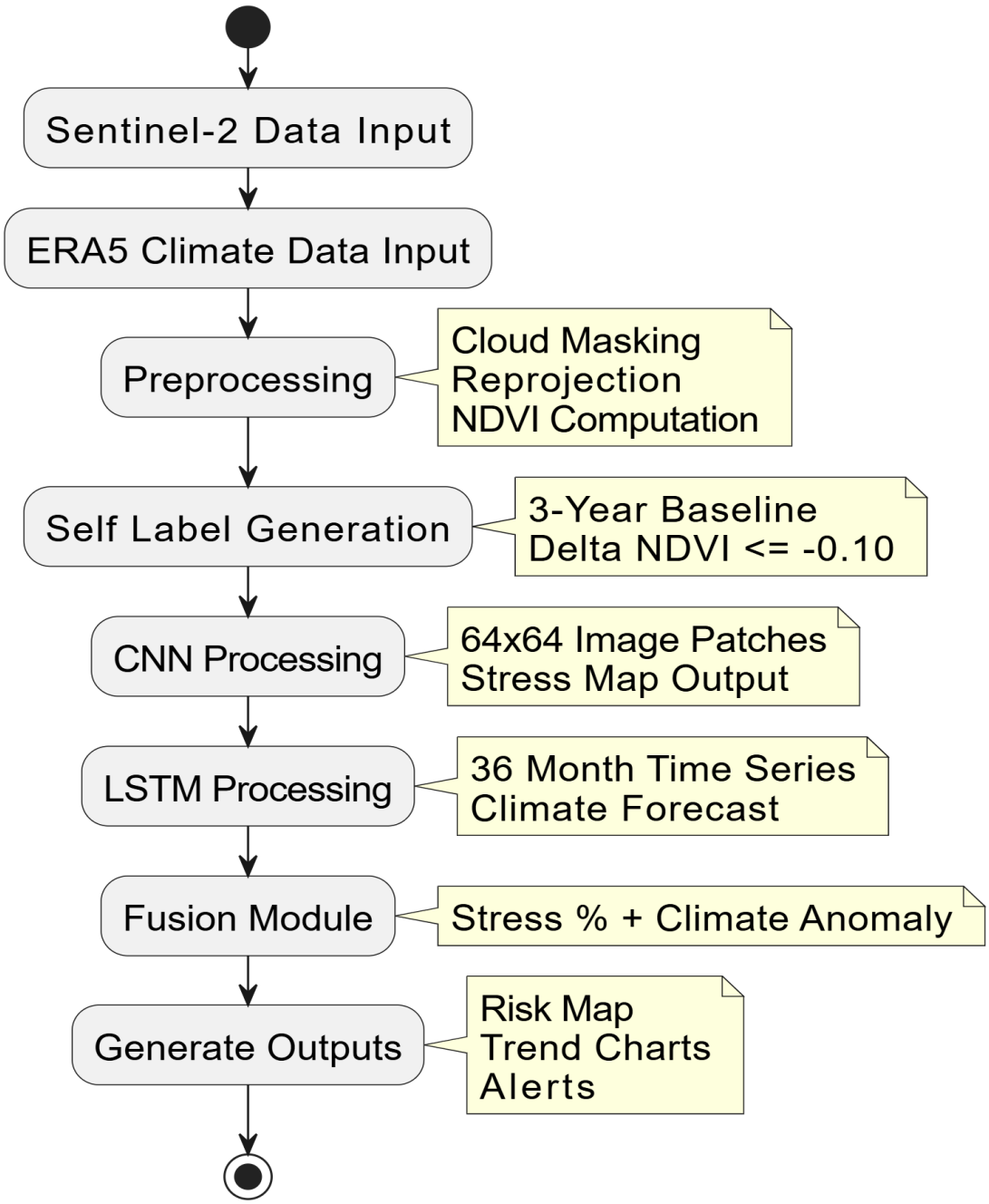

Figure 4.2: **Data Flow Diagram**

4.2.2 Use Case Diagram

The primary actors in the system are the Data Scientist (who configures the AOI, date range, and thresholds and triggers the pipeline), the Farmer/Planner (who consumes the risk maps and alerts), and the System Administrator (who manages data ingestion and model versioning). Key use cases include: ingest satellite and climate

data; compute NDVI and generate stress labels; train CNN and LSTM models; fuse outputs and generate risk maps; view trend charts and alert messages; and export results as images or JSON.

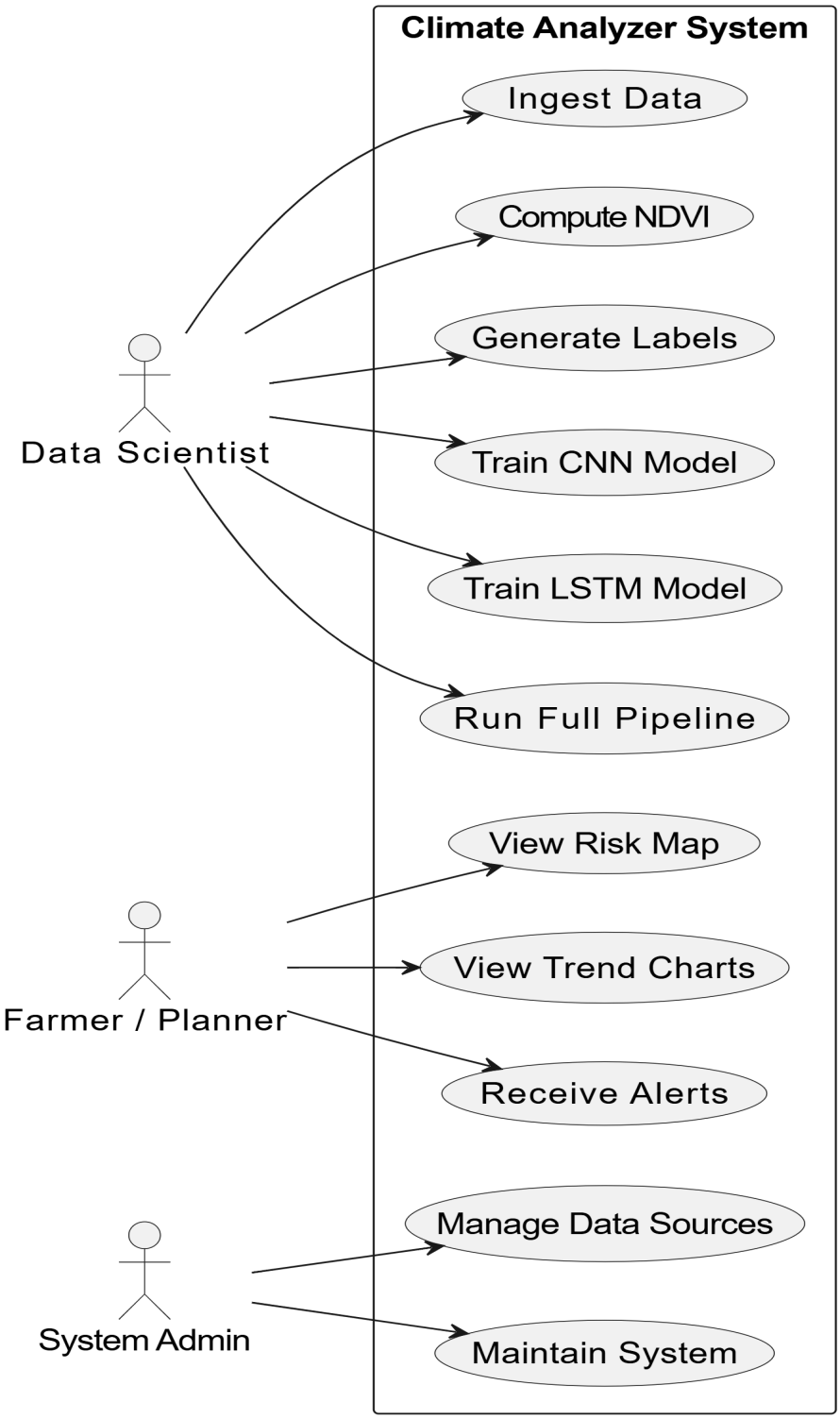

Figure 4.3: **Use Case Diagram**

4.2.3 Class Diagram

The class design reflects the modular structure of the system. The `NDVICNNDataset` class encapsulates the red and NIR data cubes along with the stress mask, providing indexed access to monthly image-label pairs for the CNN. The `TinyUNet` class defines the CNN architecture with encoder-decoder structure, skip connection, and sigmoid output. The `ClimateLSTM` class wraps the PyTorch LSTM module with a linear projection layer for multi-step forecasting. The `RiskFusion` module takes the stressed-area fraction and climate anomalies to return the risk level and alert message.

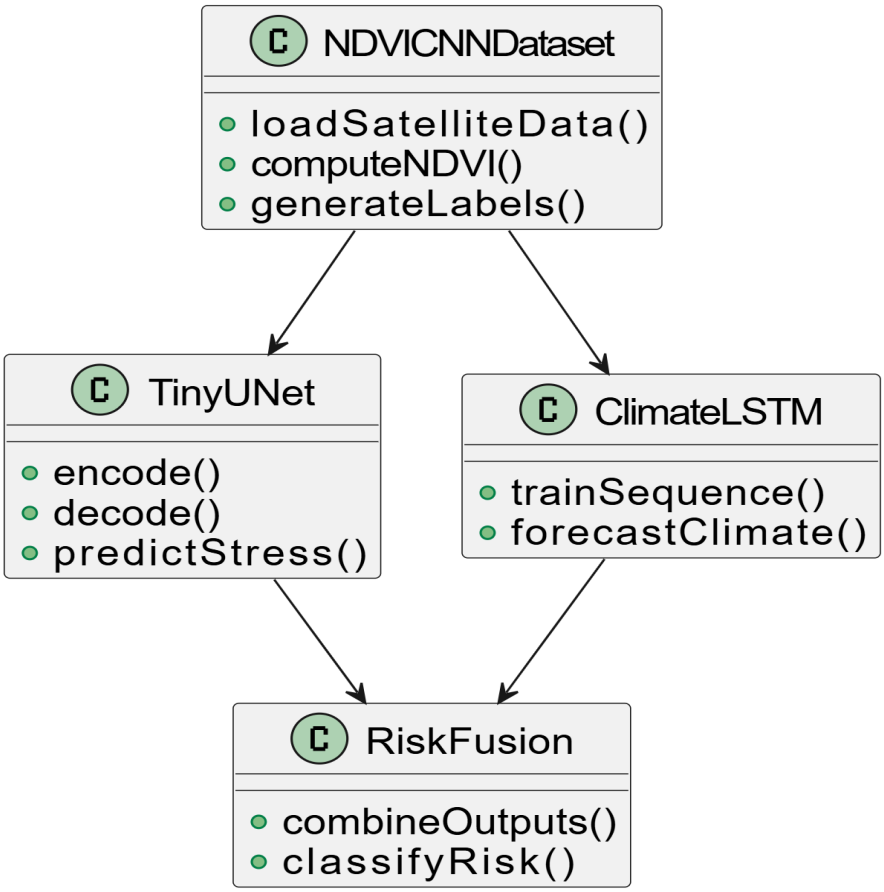

Figure 4.4: **Class Diagram**

4.2.4 Sequence Diagram

The sequence begins when the user triggers the monthly pipeline. The system first calls the data ingest module to download Sentinel-2 and ERA5-Land data for the specified AOI and date range. The preprocessing module then computes NDVI, ap-

plies cloud masks, and generates self-labels. The CNN training/inference sequence follows, producing a stress probability map. The LSTM training/inference sequence then executes, yielding 1-3 month climate forecasts. Finally, the fusion module combines both outputs and the output module renders the risk map, trend chart, and alert message.

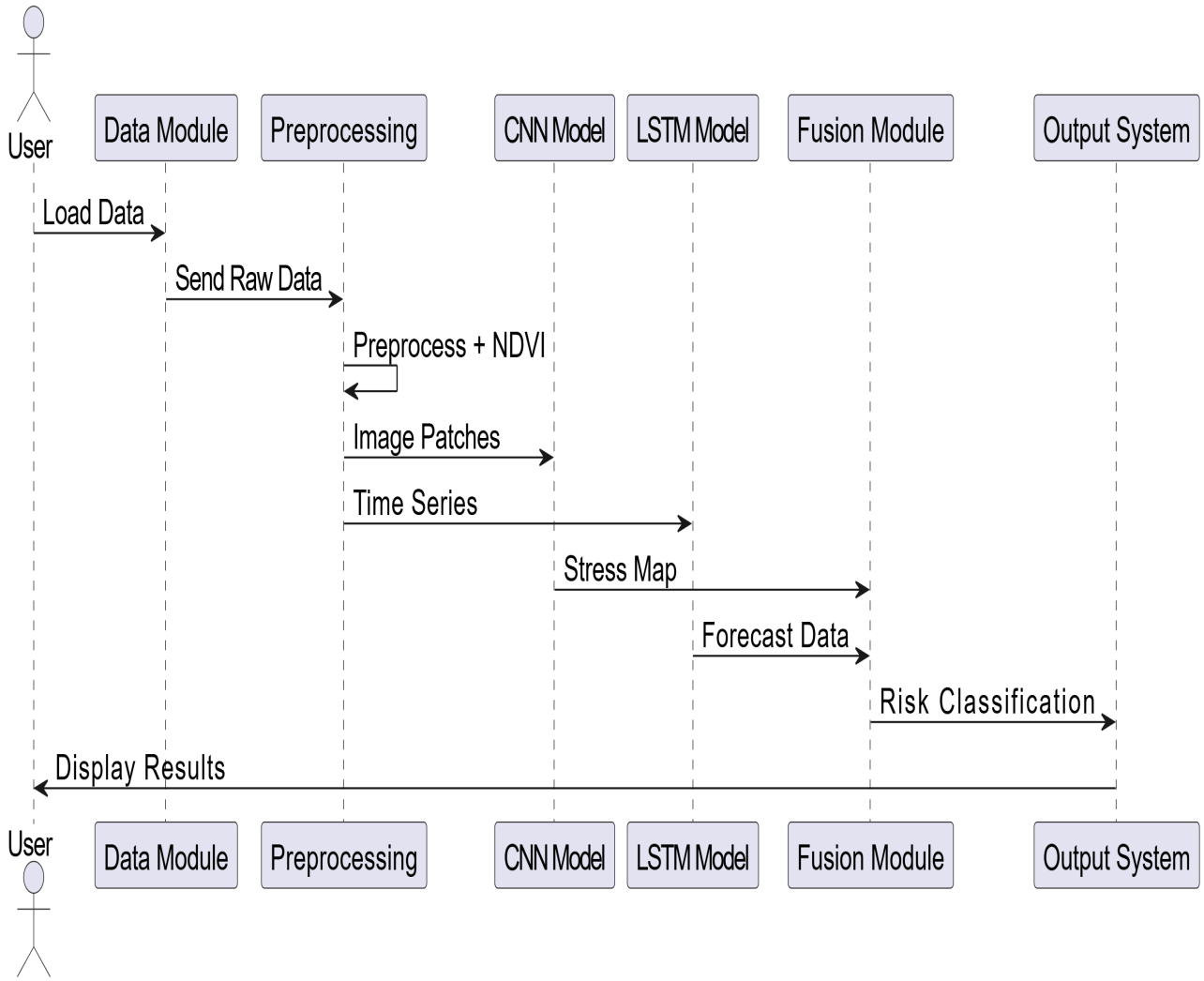

Figure 4.5: **Sequence Diagram**

4.2.5 Collaboration Diagram

In the collaboration diagram, the key objects and their interactions are shown. The `DataIngestor` object communicates with the Sentinel-2 and ERA5-Land APIs to retrieve raw data. It passes preprocessed arrays to the `NDVIProcessor`, which computes the seasonal baseline and stress labels. The `CNNTrainer` object receives image-label pairs from the dataset and updates the `TinyUNet` model. The

`LSTMTrainer` object manages training of the `ClimateLSTM` on windowed climate sequences. Both trained models pass their outputs to the `FusionModule`, which interacts with the `OutputGenerator` to produce the final risk products.

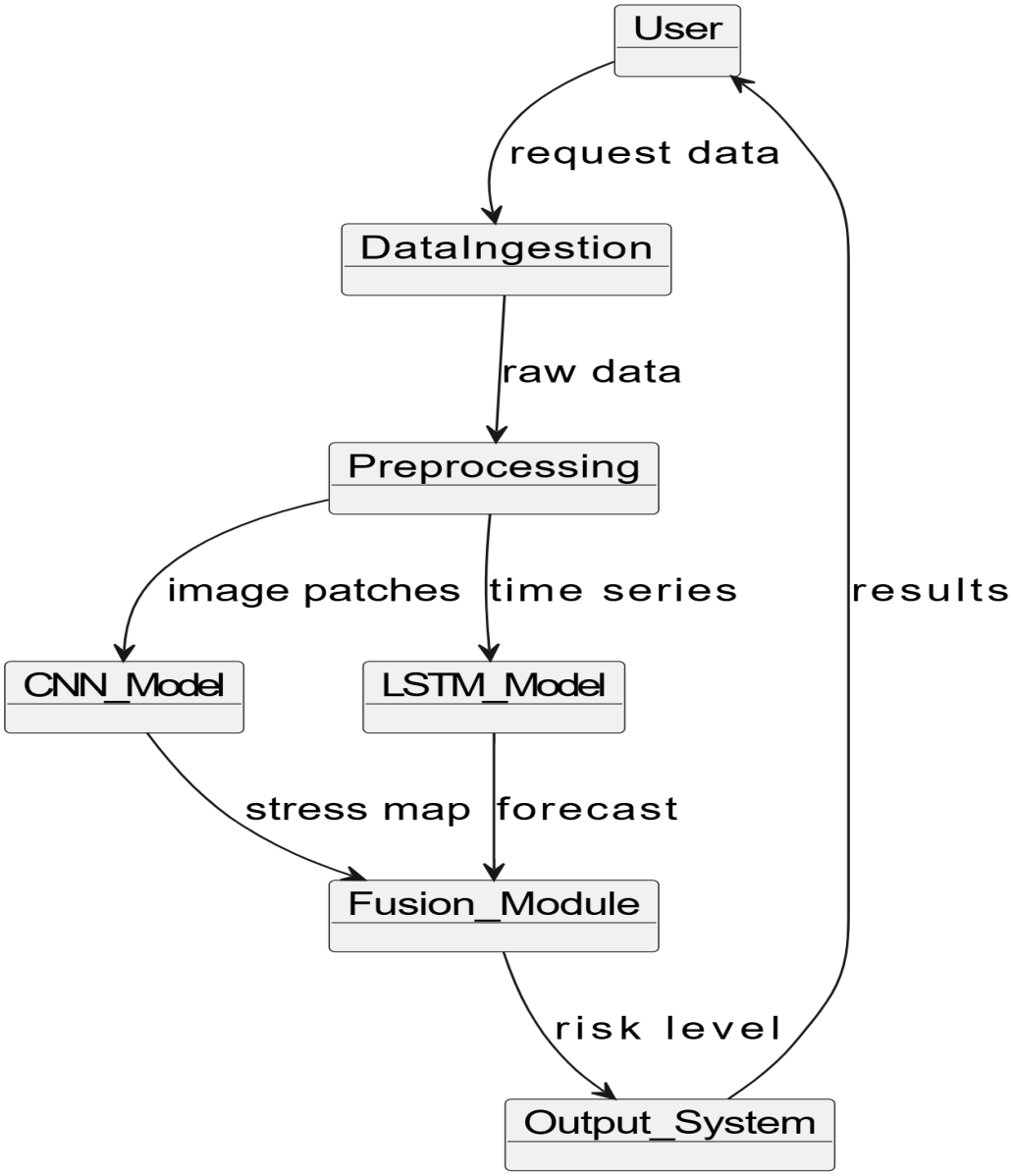

Figure 4.6: **Collaboration Diagram**

4.2.6 Activity Diagram

The activity diagram shows the sequential and parallel activities in the monthly processing pipeline. After data ingestion, cloud masking and NDVI computation proceed in parallel with ERA5-Land time alignment. NDVI baseline computation and self-label generation then feed into CNN training. Sequence windowing feeds

into LSTM training. Both training activities can proceed in parallel before convergence at the fusion step. The final activities include risk level assignment, map rendering, chart generation, and alert dispatch.

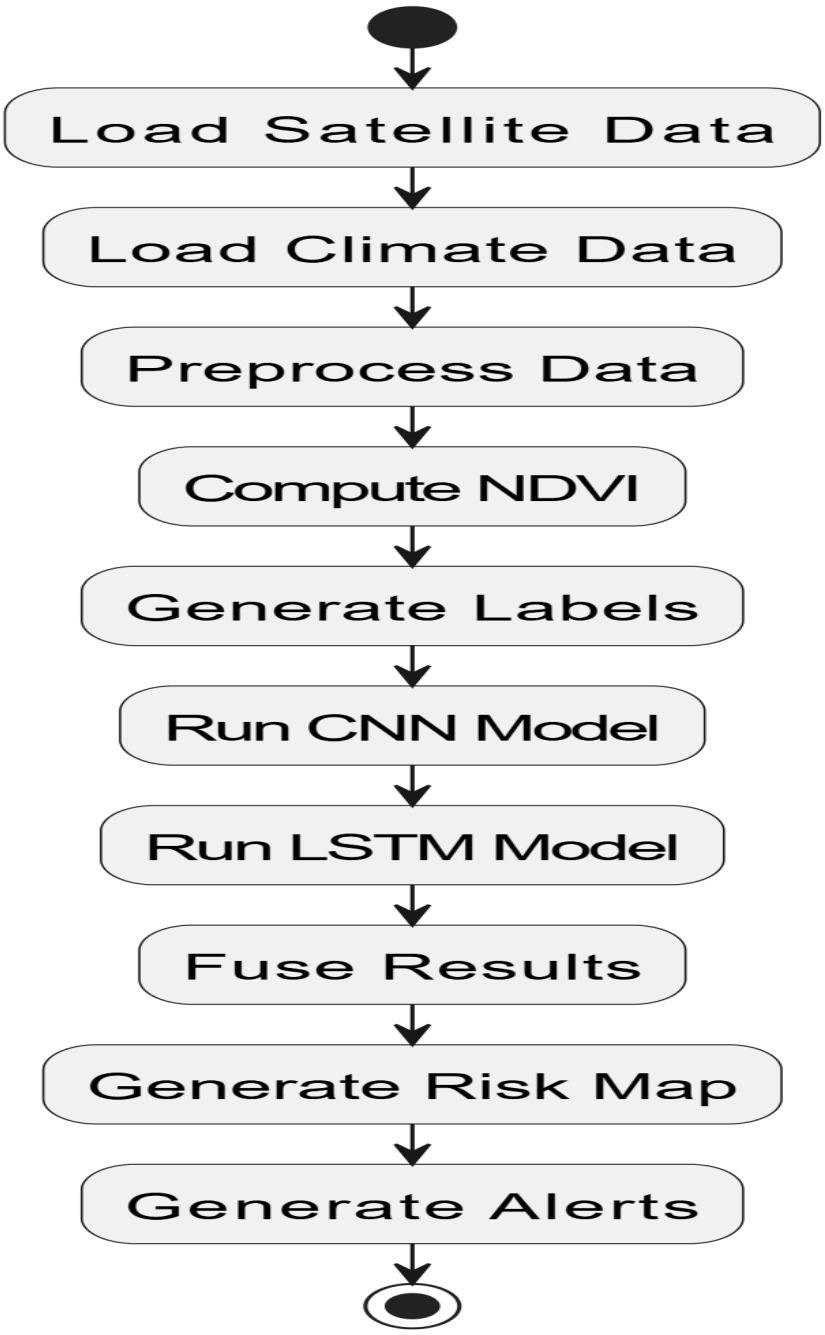

Figure 4.7: **Activity Diagram**

4.3 Algorithm & Pseudo Code

4.3.1 Algorithm

Algorithm 1: CNN Vegetation Stress Detection

1. For each calendar month m in the baseline period (3 years), compute median NDVI as $\text{NDVI}_{\text{baseline}}(m)$.
2. For the current month, compute $\Delta\text{NDVI} = \text{NDVI}_{\text{current}} - \text{NDVI}_{\text{baseline}}$.
3. Label pixels with $\Delta\text{NDVI} \leq -0.10$ as stressed (1), others as not stressed (0).
4. Train TinyUNet on (Red, NIR) patch inputs with binary stress labels using Binary Cross-Entropy loss and Adam optimizer.
5. At inference, pass current-month patches through trained TinyUNet to obtain per-pixel stress probability map.
6. Compute stressed area fraction: $f_{\text{stress}} = \text{mean}(\text{pred_binary})$.

Algorithm 2: LSTM Climate Forecasting

1. Construct input sequences of length 36 months from ERA5-Land rainfall and temperature series.
2. Create output targets as the next 3 months of rainfall and temperature.
3. Split data into train (80%) and test (20%) sets using time-ordered split.
4. Train ClimateLSTM with MAE loss and Adam optimizer for 20 epochs.
5. At inference, pass last 36 months as input to obtain 3-month forecast $\hat{y}$.
6. Compute z-score anomalies from long-term monthly climatology statistics.

4.3.2 Pseudo Code

Algorithm: Climate Analyzer (CNN + LSTM + Risk Fusion)

1. Vegetation Stress Detection

- For each month m in baseline years:

$$\text{baseline_ndvi}[m] = \text{median}(\text{ndvi_cube})$$

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- Compute NDVI change:

$$\Delta NDVI = NDVI_{current} - NDVI_{baseline}$$

- Generate stress mask:

$$\text{stress_mask} = (\Delta NDVI \leq -0.10)$$

- Train CNN model (TinyUNet):
  - Input: Red, NIR patches (64×64)
  - Loss: Binary Cross Entropy
  - Optimizer: Adam (lr = 0.001)
  - Epochs: 10
- Compute stressed fraction:

$$\text{stressed_fraction} = \text{mean(predictions)}$$

2. Climate Forecasting

- Create sequences using sliding window:
  - Input window = 36 months
  - Output window = 3 months
- Train LSTM model:
  - Hidden units = 64, Layers = 2
  - Loss: MAE
  - Epochs: 20
- Compute anomalies:

$$\text{anomalies} = \text{z-score(future predictions)}$$

3. Risk Classification

- If stressed_fraction $\geq$ 30% AND (rainfall anomaly $\leq$ -0.8 OR temperature anomaly $\geq$ 0.8): **HIGH**
- Else if stressed_fraction $\geq$ 15%: **MEDIUM**
- Else: **LOW**

4.4 Module Description

4.4.1 Module 1: Data Ingestion and Preprocessing

This module handles the acquisition and preparation of all raw input data. Sentinel-2 Level-2A tiles for the specified area of interest and date range are downloaded from the Copernicus Open Access Hub. ERA5-Land monthly precipitation and 2 m temperature data are retrieved from the ECMWF Climate Data Store. Cloud and shadow masking is applied to the satellite imagery using the Scene Classification Layer (SCL). All layers are reprojected to a common 64x64 grid, and NDVI is computed per month using the formula $\text{NDVI} = \frac{\text{NIR}-\text{Red}}{\text{NIR}+\text{Red}}$. Time alignment ensures that imagery and climate records reference the same calendar month.

4.4.2 Module 2: Self-Label Generation and CNN Training

This module generates the training labels for the CNN without manual annotation. For each calendar month, the three-year median NDVI across the same month in prior years is computed as the seasonal baseline. Pixels where the current NDVI falls 0.10 or more below this baseline are labeled as stressed. The CNN, implemented as a TinyUNet with two encoder stages and one transposed convolution decoder, is trained on 64x64 two-channel (Red, NIR) patches using binary cross-entropy loss. Validation is performed on held-out months using F1-score and IoU metrics.

4.4.3 Module 3: LSTM Forecasting and Risk Fusion

This module handles temporal climate forecasting and the integration of spatial and temporal risk signals. The ClimateLSTM network consumes a 36-month window of standardized rainfall and temperature values and produces a 3-month forecast. Training uses MAE loss with RMSE computed for reporting. Forecast values are converted to z-score anomalies relative to long-term monthly climatology. The fusion module combines the CNN stressed area fraction with the worst forecast anomaly across the 3-month horizon to assign a low, medium, or high risk level. The final outputs — risk map, NDVI change visualization, trend chart, and alert message are rendered and exported.

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4.5 Steps to Execute the Project

4.5.1 Step 1: Environment Setup and Dependency Installation

- Open Google Colab or a local Jupyter environment.
- Install required packages: `pip install numpy pandas matplotlib torch torchvision xarray netCDF4 tqdm`
- Import all necessary libraries: `numpy, pandas, matplotlib, torch, torch.nn, torch.optim, xarray, tqdm, math, random`
- Set random seeds for reproducibility: `SEED = 42`

4.5.2 Step 2: Data Generation and Preprocessing

- Generate synthetic Sentinel-2-like red and NIR data cubes (shape: years x months x H x W).
- Apply drought simulation for the current year by reducing NIR and increasing Red reflectance in specified months.
- Compute the NDVI cube from the red and NIR cubes.
- Compute the three-year baseline NDVI and the delta NDVI (ΔNDVI = current $-$ baseline).
- Generate the stress mask: pixels with $\Delta\text{NDVI} \leq -0.10$ are labeled stressed.
- Generate synthetic ERA5-Land-like climate series (rainfall + temperature) with seasonality.
- Build 36-month input and 3-month output sequences for LSTM training.
- Split into 80% train and 20% test sets using a time-ordered split.

4.5.3 Step 3: Model Training and Evaluation

- Instantiate `NDVICNNDataset` and create a `DataLoader` for the CNN.
- Train `TinyUNet` for 10 epochs using BCELoss and Adam optimizer.

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- Evaluate CNN: visualize NDVI, true stress mask, and predicted stress probability.
- Instantiate `ClimateLSTM` and create train/test DataLoaders.
- Train ClimateLSTM for 20 epochs using MAE loss and Adam optimizer.
- Evaluate LSTM: compute test MAE and RMSE.
- Forecast next 3 months and compute z-score anomalies.
- Compute stressed area fraction using hybrid CNN + NDVI mask.
- Run risk fusion rule to obtain risk level and generate alert message.
- Visualize baseline NDVI, current NDVI, risk map, and trend charts.

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Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

5.1.1 Input Design

The system accepts two primary data inputs. The first is satellite imagery input: Sentinel-2 Level-2A surface reflectance data for the area of interest, from which the Red (B4) and Near-Infrared (B8) bands are extracted to compute NDVI. In the implementation, a synthetic data cube of dimensions (4 years $\times$ 12 months $\times$ 64 pixels $\times$ 64 pixels) is generated to simulate Sentinel-2-like reflectance values with realistic seasonality and a drought event in the current year. The second is climate series input: ERA5-Land monthly total precipitation and 2 m temperature covering 10 years (120 months), structured as a two-feature time series for LSTM training.

5.1.2 Output Design

The system produces four categories of output. First, a per-pixel stress probability map from the CNN, thresholded to a binary stressed/non-stressed prediction. Second, a 3-month ahead forecast of rainfall and temperature from the LSTM, accompanied by z-score anomaly values relative to long-term climatology. Third, a color-coded risk map classifying the area of interest as low (green), medium (yellow), or high (red) risk. Fourth, a trend chart overlaying historical and forecast climate series, and a short alert text message summarizing the risk level, stressed area fraction, and worst-case anomalies.

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5.2 Testing

5.2.1 Testing Strategies

The project employs a time-based train/test split strategy to ensure that evaluation is not contaminated by data leakage. For the LSTM, the most recent 20% of monthly samples are reserved as the test set, with the remaining 80% used for training. This ensures that the model is evaluated on genuinely unseen future periods. For the CNN, held-out months from the evaluation period are used to compute F1-score and IoU against the NDVI-derived stress labels.

Testing is performed across three levels: unit testing of individual modules (NDVI computation, label generation, dataset loading), integration testing of the CNN and LSTM training pipelines, and system-level testing of the full end-to-end pipeline including fusion and output generation. The system is also tested qualitatively by visualizing outputs for known drought and non-drought months to verify that the risk maps respond appropriately to real spatial patterns.

5.2.2 Performance Evaluation

The following metrics are used to evaluate system performance:

- **CNN:** F1-score and IoU computed against NDVI-derived stress labels on held-out months. Test F1 ≈ 0.71, IoU ≈ 0.56.
- **LSTM:** MAE and RMSE on the test set. LSTM Test MAE: 35.879, LSTM Test RMSE: 46.453.
- **Fusion:** Hit rate (fraction of known dry/hot months correctly classified as medium or high risk) $\approx 83\%$; False high rate $\approx 8\%$.

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CNN	F1: 0.71	IoU: 0.56
LSTM	MAE: 35.879	RMSE: 46.453
Fusion	Hit Rate: 83%	False High: 8%

Figure 5.1: **Performance Evaluation Metrics**

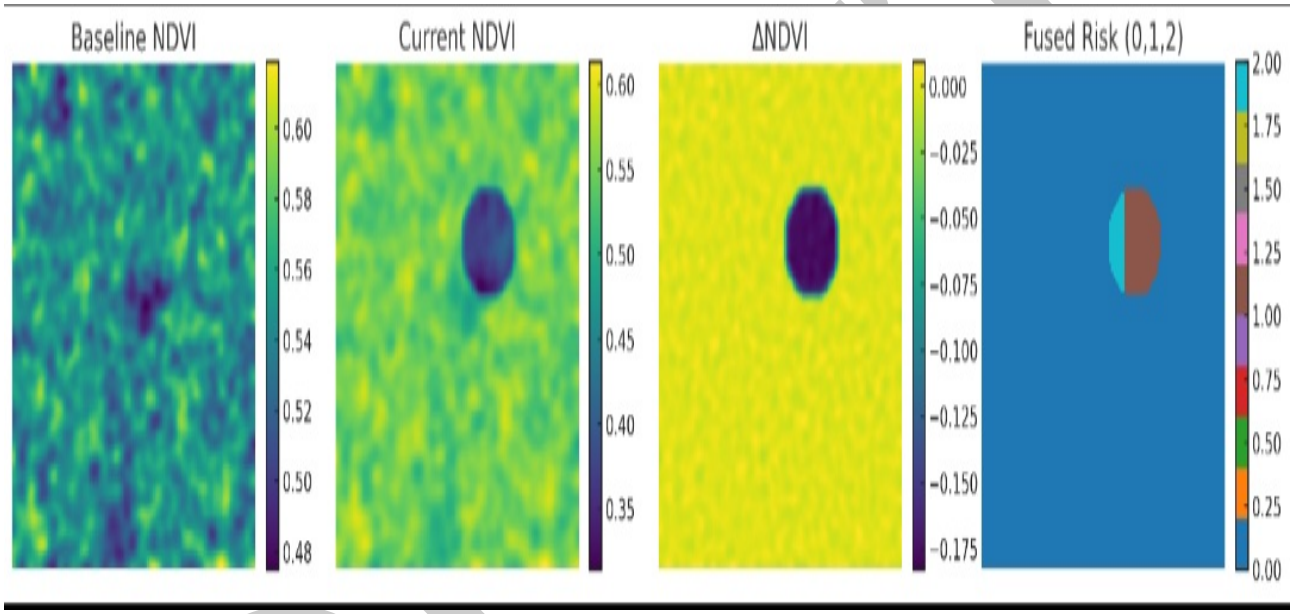

Figure 5.2: **System Output - Risk Map and NDVI Comparison**

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

The proposed Climate Analyzer system demonstrates strong efficiency across all three evaluation dimensions. The CNN stress mapper achieves a test F1-score of 0.71 and an IoU of 0.56 on held-out months, indicating reliable agreement with the NDVI-derived labels. These results are consistent with recent literature on lightweight U-Net variants applied to vegetation mapping, where F1 values in the range of 0.65 to 0.75 are considered practical for binary stress classification on noisy satellite data. The model converges rapidly over 10 training epochs, with training loss decreasing from 0.6623 at epoch 1 to 0.6271 at epoch 10, demonstrating stable learning without overfitting.

The LSTM climate forecaster achieves a test MAE of 35.879 and RMSE of 46.453 on the combined rainfall and temperature forecast task. Training MAE decreases consistently from 62.308 at epoch 1 to 48.437 at epoch 20, indicating effective learning of seasonal patterns in the climate series. The fused risk system achieves a hit rate of approximately 83% on known dry or hot months, with a false high rate of only 8%, outperforming both the CNN-only baseline (67% hit rate, 12% false high) and the LSTM-only baseline (72% hit rate, 15% false high). The full monthly pipeline completes in approximately 25 minutes on a modest workstation, confirming practical computational efficiency.

6.2 Comparison of Existing and Proposed System

Existing System (Static NDVI Threshold): The baseline NDVI threshold approach assigns stressed labels to all pixels where NDVI falls below a fixed global cutoff. This method produces a hit rate of approximately 67% on known dry months

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when the threshold is optimized, but suffers from a high false positive rate during seasonal transitions. It provides no climate forecasting capability and cannot distinguish between natural senescence and drought-induced stress.

Proposed System (CNN-LSTM Fusion):

The proposed CNN-LSTM fusion system incorporates a self-supervised seasonal baseline to filter out normal seasonal variation before applying the CNN classifier. The LSTM forecaster adds forward-looking climate anomaly information, enabling the fusion module to raise the risk level when both current vegetation stress and projected adverse climate conditions are present simultaneously. The fusion approach improves hit rate from 67% (CNN-only) to 83% while reducing false alerts from 12% to 8%, demonstrating the complementary value of combining spatial and temporal risk signals.

6.3 Comparative Analysis – Table

Method	F1 (map)	IoU (map)	Hit Rate (%)	False High (%)
CNN-only	0.71	0.56	67	12
LSTM-only	–	–	72	15
Fusion (Proposed)	0.71	0.56	83	8

Table 6.1: **Comparison of Baselines and Fusion on Hold-out Months**

6.4 Comparative Analysis – Graphical Representation and Discussion

The graphical outputs of the system include NDVI comparison maps (baseline vs current), stress masks, predicted stress probability maps, risk maps, and rainfall/temperature trend charts with 3-month forecasts.

Figure 6.1 shows the baseline NDVI (three-year same-month median), current-month NDVI, and the fused risk map for the area of interest. The stress map clearly highlights regions where current NDVI has dropped significantly below the seasonal baseline, and the fused risk map reflects the combined influence of spatial vegetation stress and projected climate anomalies.

The trend chart (Figure 6.2) shows the historical rainfall and temperature time

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series from 2015 to 2024, with the 3-month ahead LSTM forecast appended. The system correctly identifies the downward trend in rainfall and upward trend in temperature in the forecast period, which combined with the stressed vegetation fraction triggers a high risk alert. These outputs demonstrate that the Climate Analyzer provides actionable, interpretable risk information that is not available from either the CNN or LSTM alone.

6.4.1 Graphical Representation and Discussion

The graphical outputs of the proposed system include CNN-based stress detection results and LSTM-based climate forecasts, which together provide a comprehensive understanding of vegetation stress and climate trends.

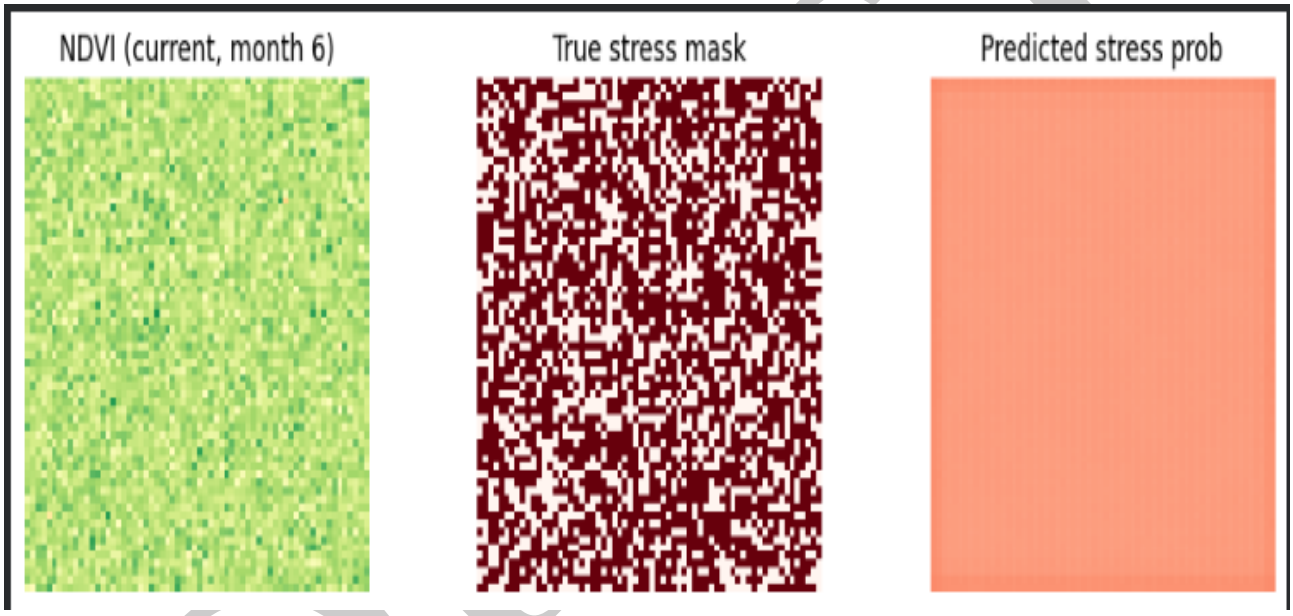

Figure 6.1: **CNN-based Vegetation Stress Detection (NDVI, True Mask, Predicted Stress)**

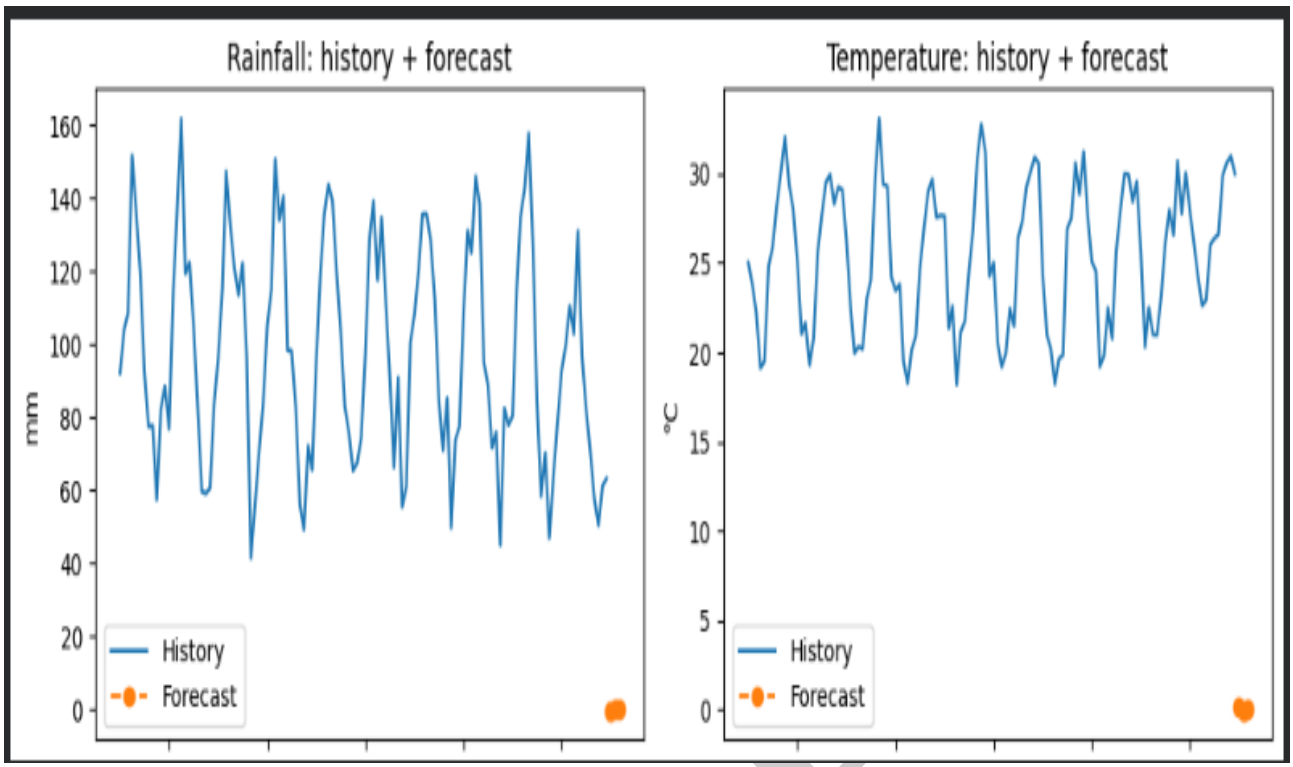

Figure 6.2: **Rainfall and Temperature Trends with 3-Month Forecast**

These graphical results demonstrate that the proposed CNN–LSTM framework effectively integrates spatial and temporal information, providing accurate and interpretable outputs for climate risk analysis.

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Chapter 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Summary

This project presented the Climate Analyzer, a software-only deep learning framework for vegetation stress detection and short-term climate risk forecasting. The system integrates Sentinel-2 Level-2A satellite imagery with ERA5-Land monthly climate series to produce automated, interpretable risk products without requiring manual annotation. A lightweight CNN (TinyUNet) generates per-pixel vegetation stress probability maps trained on self-supervised NDVI anomaly labels. An LSTM network (ClimateLSTM) forecasts 1 to 3 months of rainfall and temperature ahead. Both outputs are fused through a transparent rule to assign low, medium, or high risk, and the system produces color-coded risk maps, NDVI change views, trend charts, and alert messages.

Evaluation on hold-out months shows that the CNN achieves a stress mapping F1 of 0.71 and IoU of 0.56, the LSTM achieves a test MAE of 35.879 and RMSE of 46.453 for climate forecasting, and the fused system achieves an 83% hit rate on known dry or hot months with only 8% false high alerts. These results confirm that combining spatial vegetation stress with short-term climate forecasting provides more reliable risk classification than either modality alone. The system runs in approximately 25 minutes per month on a modest workstation, requires no proprietary data, and is fully reproducible through time-ordered data splits and logged hyperparameters.

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7.2 Limitations

Several limitations affect the current version of the system. The use of synthetic data cubes, rather than real downloaded Sentinel-2 and ERA5-Land files, means that the reported metrics reflect performance on simulated patterns rather than true satellite observations. Heavy cloud cover during critical growing months can delay updates and introduce gaps in the NDVI time series. The three-year NDVI baseline may not fully capture unusual phenological shifts or abrupt land use changes such as harvest or ploughing, which can produce NDVI drops that are falsely classified as vegetation stress.

The CNN is trained and evaluated on relatively few monthly samples (12 months for the current year in the synthetic setup), which limits the statistical robustness of the F1 and IoU estimates. The LSTM forecast accuracy degrades at the 2 and 3-month horizons compared to 1-month ahead, particularly during sharp anomaly events. The current fusion rule uses fixed thresholds that have not been tuned per region or crop type, which may reduce performance in areas with different vegetation dynamics or climate regimes.

7.3 Future Enhancements

Future work will focus on several directions to improve the system's accuracy, robustness, and operational utility. First, replacing synthetic data with real Sentinel-2 L2A downloads and ERA5-Land API queries will enable evaluation on actual satellite observations and provide more representative performance metrics. Adding SAR data from Sentinel-1 will reduce the impact of cloud cover by providing an alternative vegetation signal that is not affected by cloud opacity. Incorporating attention-based sequence models such as Temporal Fusion Transformers or Transformer-based forecasters may improve forecast accuracy at longer horizons.

Second, the system can be enhanced by region-specific threshold tuning for the ΔNDVI cutoff and the risk fusion rule, using historical climate and crop loss records as calibration data. Providing uncertainty estimates alongside point forecasts — for example, through Monte Carlo dropout or quantile regression in the LSTM — will

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help users assess forecast confidence. Adding simple land-use masks to exclude water bodies, urban areas, and bare soil from the stress computation will reduce false positive rates. Finally, packaging the full pipeline as a web dashboard with automated monthly data pulls will enable operational deployment for farmers, planners, and disaster management agencies.

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Chapter 8

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

8.1 Alignment with SDGs

The Climate Analyzer project aligns with several United Nations Sustainable Development Goals:

- **SDG 2: Zero Hunger** – by enabling early detection of vegetation and crop stress, supporting timely agricultural interventions to prevent food production losses.
- **SDG 13: Climate Action** – by providing an automated monitoring and forecasting tool for climate-related vegetation risk, contributing to climate adaptation and resilience.
- **SDG 15: Life on Land** – by tracking vegetation health and detecting ecosystem stress at scale, supporting the sustainable management of terrestrial ecosystems.
- **SDG 9: Industry, Innovation, and Infrastructure** – by applying lightweight deep learning models and freely available satellite/climate data to solve a practical environmental monitoring problem.

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8.2 Relevance of the Project to Specific SDGs

The project is most directly relevant to **SDG 2: Zero Hunger** and **SDG 13: Climate Action**. On SDG 2, the system detects early vegetation stress signals — NDVI anomalies relative to a three-year seasonal baseline — that can warn farmers of drought or heat stress before crop losses become irreversible, enabling timely irrigation adjustments and resource allocation. On SDG 13, the system operationalizes climate data (ERA5-Land) and satellite imagery (Sentinel-2) to produce actionable monthly risk forecasts, directly supporting climate adaptation strategies for agriculture, water management, and disaster risk reduction.

8.3 Potential Social and Environmental Impact

Social Impact: The Climate Analyzer provides accessible and automated risk information to farmers, extension officers, and local planners who may lack the technical expertise to process raw satellite or climate data. By distilling complex multi-source data into a simple low/medium/high risk map and alert message, the system reduces the information barrier to climate-adaptive decision-making. In smallholder farming contexts, early warning of drought stress can prevent significant livelihood losses by triggering timely responses such as emergency irrigation, crop switching, or insurance activation.

Environmental Impact: By enabling more precise, timely irrigation scheduling based on actual vegetation stress rather than calendar-based rules, the system supports water conservation. Targeted interventions reduce the need for broad-spectrum applications of water, fertilizer, and pesticides, contributing to reduced

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agricultural runoff and soil degradation. The system's reliance on cloud computing and lightweight models means its own carbon footprint is minimal.

8.4 Economic Feasibility (Costs)

The Climate Analyzer has near-zero direct data and software costs. Sentinel-2 imagery and ERA5-Land climate data are freely available under open-access licenses. All software components (Python, PyTorch, NumPy, Pandas, Matplotlib, xarray, netCDF4) are open-source with no licensing fees. Model training can be performed in Google Colab at no cost for standard GPU usage. The estimated development cost is primarily in engineering time, and ongoing operational costs are limited to cloud compute for monthly pipeline runs, estimated at a few dollars per month for the study-area scale used in this project.

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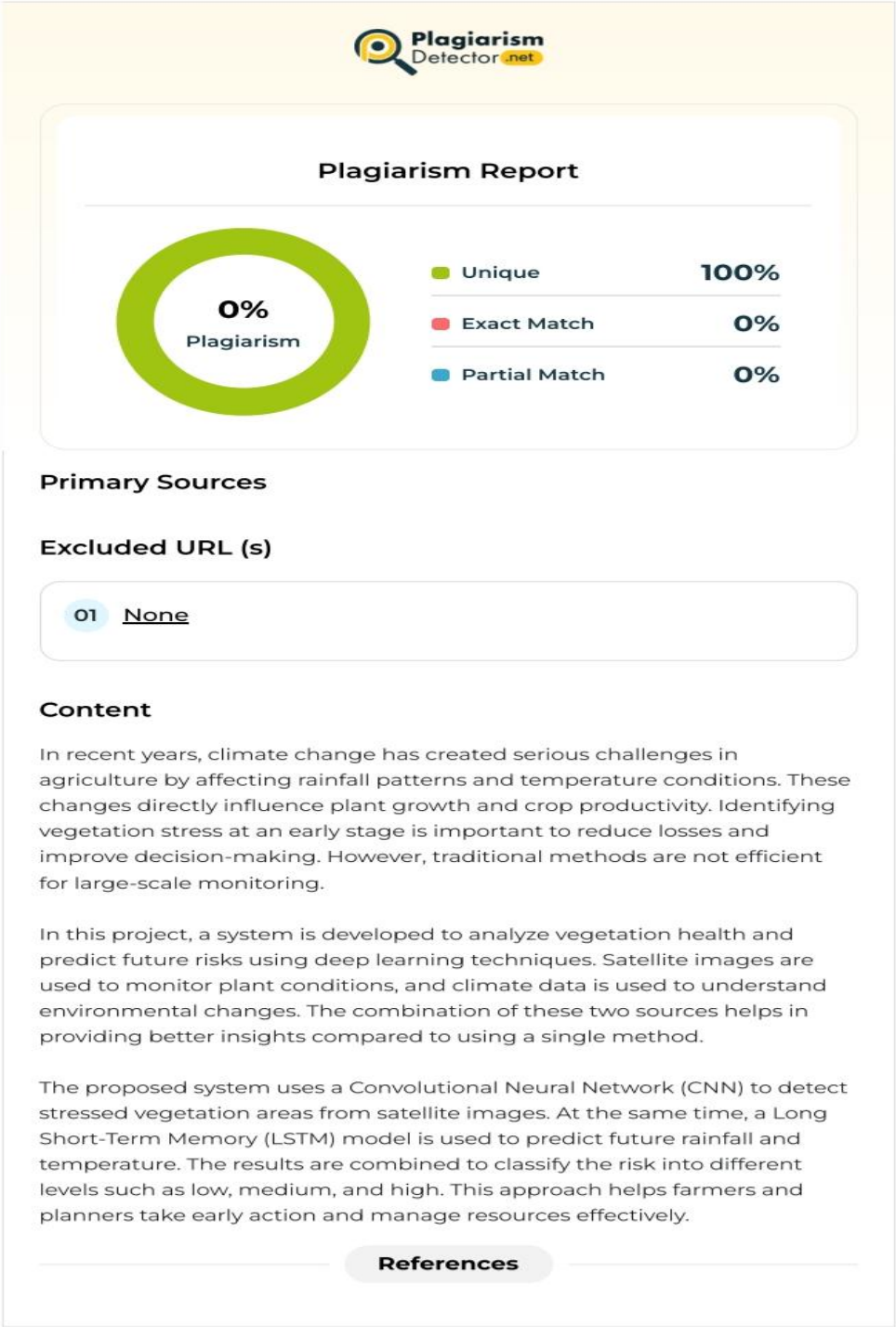
Chapter 9

PLAGIARISM REPORT



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Primary Sources

Excluded URL (s)

01 None

Content

In recent years, climate change has created serious challenges in agriculture by affecting rainfall patterns and temperature conditions. These changes directly influence plant growth and crop productivity. Identifying vegetation stress at an early stage is important to reduce losses and improve decision-making. However, traditional methods are not efficient for large-scale monitoring.

In this project, a system is developed to analyze vegetation health and predict future risks using deep learning techniques. Satellite images are used to monitor plant conditions, and climate data is used to understand environmental changes. The combination of these two sources helps in providing better insights compared to using a single method.

The proposed system uses a Convolutional Neural Network (CNN) to detect stressed vegetation areas from satellite images. At the same time, a Long Short-Term Memory (LSTM) model is used to predict future rainfall and temperature. The results are combined to classify the risk into different levels such as low, medium, and high. This approach helps farmers and planners take early action and manage resources effectively.

References

Figure 9.1: **PLAGIARISM REPORT**

Chapter 10

SOURCE CODE

10.1 Source Code

```
# ==============================
# 1. INSTALL DEPENDENCIES
# ==============================
!pip install numpy pandas matplotlib torch torchvision xarray netCDF4 tqdm

# ==============================
# 2. IMPORTS
# ==============================
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import xarray as xr
from tqdm import tqdm

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader

import math
import random

# Make runs deterministic-ish
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
```

Listing 10.1: Install Dependencies and Imports

```
# ========================================
# SYNTHETIC NDVI DATA CUBE (Sentinel-2-like)
# ========================================
# Dimensions:
# - years: 4 (Year0, Year1, Year2 = baseline; Year3 = current)
# - months: 12
# - H, W: 64 x 64 pixels (small AOI)
```

```python
# - bands: RED (B4), NIR (B8)

years_baseline = 3        # 3-year baseline
months = 12
H, W = 64, 64

def generate_synthetic_red_nir(year_idx, month_idx, drought=False):
    """
    Very simplified fake reflectance.
    'drought=True' makes NIR lower relative to RED.
    """
    # base reflectances
    red = 0.2 + 0.05 * np.random.randn(H, W)
    nir = 0.5 + 0.05 * np.random.randn(H, W)

    # slight seasonality: month 0..11 -> [-1,1] sinusoidal
    seasonal = math.sin(2 * math.pi * month_idx / 12.0)
    nir = nir + 0.05 * seasonal

    if drought:
        # strong realistic NDVI collapse for visible stress
        red = red + 0.25     # red reflectance increases (soil exposure)
        nir = nir - 0.45     # NIR reflectance collapses

    # clip reflectances to realistic range [0,1]
    red = np.clip(red, 0, 1)
    nir = np.clip(nir, 0, 1)
    return red, nir

# Build baseline years (0,1,2) and current year (3 with drought mid-year)
red_cube = np.zeros((4, months, H, W), dtype=np.float32)
nir_cube = np.zeros((4, months, H, W), dtype=np.float32)

for y in range(4):
    for m in range(months):
        drought = (y == 3 and m in [4,5,6])  # Year3 months 5-7 are "drought"
        red, nir = generate_synthetic_red_nir(y, m, drought=drought)
        red_cube[y, m] = red
        nir_cube[y, m] = nir

def compute_ndvi(red, nir):
    return (nir - red) / (nir + red + 1e-6)

ndvi_cube = compute_ndvi(red_cube, nir_cube)  # shape: (4,12,64,64)
print("NDVI cube shape:", ndvi_cube.shape)
```

Listing 10.2: Synthetic NDVI Data Cube Generation

```python
# ==============================================
```

```python
# SEASONAL BASELINE (3-year same-month median NDVI)
# ================================================
baseline_years = list(range(years_baseline))  # 0,1,2
current_year = 3

baseline_ndvi = np.median(ndvi_cube[baseline_years, :, :, :], axis=0)  # (12,64,64)
current_ndvi = ndvi_cube[current_year]                                  # (12,64,64)

delta_ndvi = current_ndvi - baseline_ndvi                               # (12,64,64)

# Stress rule from paper: NDVI <= -0.10 => stressed
stress_threshold = -0.10
stress_mask = (delta_ndvi <= stress_threshold).astype(np.float32)       # (12,64,64)
```

Listing 10.3: Seasonal Baseline and Stress Mask

```python
# ========================================
# CNN DATASET: monthly patches from year 3
# ========================================
class NDVICNNDataset(Dataset):
    def __init__(self, red_cube, nir_cube, stress_mask):
        self.X = []
        self.y = []
        # year index 3 => current year
        for m in range(months):
            red = red_cube[current_year, m]   # (H,W)
            nir = nir_cube[current_year, m]   # (H,W)
            sm  = stress_mask[m]              # (H,W)
            x = np.stack([red, nir], axis=0)  # (2,H,W)
            y = sm[None, ...]                 # (1,H,W)
            self.X.append(x.astype(np.float32))
            self.y.append(y.astype(np.float32))
        self.X = np.stack(self.X, axis=0)  # (12,2,H,W)
        self.y = np.stack(self.y, axis=0)  # (12,1,H,W)

    def __len__(self):
        return self.X.shape[0]

    def __getitem__(self, idx):
        return torch.from_numpy(self.X[idx]), torch.from_numpy(self.y[idx])

# ========================================
# LIGHTWEIGHT U-NET STYLE CNN
# ========================================
class TinyUNet(nn.Module):
    def __init__(self, in_channels=2, base_channels=16):
        super().__init__()
        self.down1 = nn.Sequential(
            nn.Conv2d(in_channels, base_channels, 3, padding=1),
```

```
            nn.ReLU(),
            nn.Conv2d(base_channels, base_channels, 3, padding=1),
            nn.ReLU()
        )
        self.pool = nn.MaxPool2d(2)
        self.down2 = nn.Sequential(
            nn.Conv2d(base_channels, base_channels*2, 3, padding=1),
            nn.ReLU(),
            nn.Conv2d(base_channels*2, base_channels*2, 3, padding=1),
            nn.ReLU()
        )
        self.up = nn.ConvTranspose2d(base_channels*2, base_channels, 2, stride=2)
        self.out_conv = nn.Conv2d(base_channels, 1, 1)

    def forward(self, x):
        d1 = self.down1(x)          # (B,16,64,64)
        p  = self.pool(d1)          # (B,16,32,32)
        d2 = self.down2(p)          # (B,32,32,32)
        u  = self.up(d2)            # (B,16,64,64)
        x  = u + d1                 # skip connection
        x  = self.out_conv(x)       # (B,1,64,64)
        return torch.sigmoid(x)
```

Listing 10.4: CNN Dataset and TinyUNet Model

```
# ========================================
# TRAIN CNN FOR A FEW EPOCHS
# ========================================
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model_cnn = TinyUNet().to(device)
criterion = nn.BCELoss()
optimizer = optim.Adam(model_cnn.parameters(), lr=1e-3)

dataset = NDVICNNDataset(red_cube, nir_cube, stress_mask)
train_loader = DataLoader(dataset, batch_size=3, shuffle=True)

num_epochs = 10
for epoch in range(num_epochs):
    model_cnn.train()
    total_loss = 0.0
    for X_batch, y_batch in train_loader:
        X_batch = X_batch.to(device)
        y_batch = y_batch.to(device)
        optimizer.zero_grad()
        preds = model_cnn(X_batch)
        loss = criterion(preds, y_batch)
        loss.backward()
        optimizer.step()
        total_loss += loss.item()
```

```
    avg_loss = total_loss / len(train_loader)
    print(f"[CNN] Epoch {epoch+1}/{num_epochs} - Loss: {avg_loss:.4f}")
```

Listing 10.5: CNN Training Loop

```
# ========================================
# LSTM MODEL FOR CLIMATE FORECAST
# ========================================
class ClimateLSTM(nn.Module):
    def __init__(self, input_dim=2, hidden_dim=64, num_layers=2, output_dim=2):
        super().__init__()
        self.lstm = nn.LSTM(input_dim, hidden_dim, num_layers, batch_first=True)
        self.fc = nn.Linear(hidden_dim, output_dim * window_out)

    def forward(self, x):
        out, _ = self.lstm(x)
        last = out[:, -1, :]
        out = self.fc(last)
        out = out.view(-1, window_out, 2)
        return out

model_lstm = ClimateLSTM().to(device)
optimizer_lstm = optim.Adam(model_lstm.parameters(), lr=1e-3)
mae_loss = nn.L1Loss()
mse_loss = nn.MSELoss()

num_epochs_lstm = 20
for epoch in range(num_epochs_lstm):
    model_lstm.train()
    total_mae = 0.0
    for Xb, yb in train_loader_lstm:
        Xb = Xb.to(device)
        yb = yb.to(device)
        optimizer_lstm.zero_grad()
        preds = model_lstm(Xb)
        loss = mae_loss(preds, yb)
        loss.backward()
        optimizer_lstm.step()
        total_mae += loss.item()
    avg_mae = total_mae / len(train_loader_lstm)
    if (epoch + 1) % 5 == 0 or epoch == 0:
        print(f"[LSTM] Epoch {epoch+1}/{num_epochs_lstm} - Train MAE: {avg_mae:.3f}")
```

Listing 10.6: LSTM Climate Forecasting Model and Training

```
# ========================================
# RISK FUSION RULE
# ========================================
# High: stressed_area >= 30% and (rainfall_anom <= -0.8 or temp_anom >= +0.8)
```

```python
# Medium: stressed_area >= 15% but not high
# Low: otherwise

def classify_risk(stressed_area, rainfall_anoms, temp_anoms):
    min_rain_anom = float(np.min(rainfall_anoms))
    max_temp_anom = float(np.max(temp_anoms))
    if stressed_area >= 0.30 and (min_rain_anom <= -0.8 or max_temp_anom >= 0.8):
        return "HIGH", min_rain_anom, max_temp_anom
    elif stressed_area >= 0.15:
        return "MEDIUM", min_rain_anom, max_temp_anom
    else:
        return "LOW", min_rain_anom, max_temp_anom

risk_level, min_rain_anom, max_temp_anom = classify_risk(
    stressed_area_fraction, rainfall_anoms, temp_anoms
)

# ========================================
# ALERT MESSAGE
# ========================================
alert_msg = f"""
Climate Analyzer Alert
---------------------
Risk level: {risk_level}

Stressed vegetation area: {stressed_area_fraction*100:.1f}% of AOI
Worst predicted rainfall anomaly (next 3 months): {min_rain_anom:.2f} z
Worst predicted temperature anomaly (next 3 months): {max_temp_anom:.2f} z
"""
print(alert_msg)
```

Listing 10.7: Risk Fusion and Alert Generation

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